ANAND KUMAR SHRIVASTAV

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and Generative AI Engineer with 3.5+ years of experience designing and deploying Machine Learning and Retrieval-Augmented Generation (RAG) solutions across healthcare, financial analytics, and digital payment fraud detection systems. Strong expertise in LLMs, semantic search, embeddings, vector databases, predictive modeling, and anomaly detection to build intelligent decision-support systems. Experienced in developing end-to-end ML and GenAI pipelines including data ingestion, EDA, feature engineering, model training, evaluation, and production deployment using Python-based microservices. Proven ability to deliver scalable AI solutions that improve operational efficiency, enhance decision-making, and reduce manual effort through FastAPI, Docker, and AWS-based cloud deployments.

TECHNICAL SKILLS

Generative AI & NLP: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Transformers, Semantic Search, Embeddings, Vector Databases (Pinecone)

Machine Learning & Data Science: Machine Learning, Predictive Modeling, Anomaly Detection, Feature Engineering, Exploratory Data Analysis (EDA), Model Evaluation, Model Optimization

Algorithms & Frameworks: Logistic Regression, Random Forest, XGBoost, Isolation Forest, Deep Learning, TensorFlow, PyTorch, Scikit-learn

Programming & Data Tools: Python, SQL, Pandas, NumPy, MySQL, MongoDB

Deployment & MLOps: FastAPI, REST APIs, Docker, AWS Cloud, CI/CD, Model Serving, Microservices Architecture, Model Monitoring

Data Visualization: Matplotlib, Seaborn, Power BI, Tableau

PROFESSIONAL EXPERIENCE

Data Scientist Oct 2022 – Present
TECEON Software Pvt.Ltd., New Delhi

- Develop and deploy enterprise-scale Machine Learning and Generative AI solutions across healthcare analytics, financial intelligence systems, and digital payment fraud detection using Python, NLP, and predictive modeling techniques.
- Design and implement Retrieval-Augmented Generation (RAG) architectures using Large Language Models, embeddings, semantic search, and vector databases to build intelligent document retrieval and decisionsupport systems.
- Build end-to-end data science pipelines including data preprocessing, exploratory data analysis (EDA), feature engineering, model training, and evaluation using algorithms such as Logistic Regression, Random Forest, XGBoost, and Isolation Forest.
- Develop scalable AI-powered APIs and microservices using FastAPI, Docker, and AWS, enabling real-time fraud risk scoring, financial insights generation, and patient clinical data summarization.

- Improve business outcomes by implementing advanced model evaluation and monitoring techniques including Precision, Recall, F1-score, AUC, KS statistics, retrieval accuracy, and latency optimization to ensure reliable and production-ready AI systems.

PROJECTS

Patient History Intelligence Engine using RAG Pipelines Project

Duration: 3 months | Team Size: 5

Description:

Developed an enterprise-grade Retrieval-Augmented Generation (RAG) based Patient History Intelligence Engine to retrieve and summarize structured EHR data and unstructured clinical documents. The solution enables accurate, context-grounded clinical insights, reduces manual chart review effort, and supports data-driven clinical decision-making in a privacy-compliant environment.

Roles And Responsibilities:

- Designed and implemented enterprise-grade RAG-based Generative AI solutions for healthcare use cases using secure, closed-source LLMs.
- Owned end-to-end GenAI and data science workflows, including data preprocessing, embedding generation, and LLM integration for clinical insights.
- Built and optimized patient-specific RAG pipelines leveraging semantic search to deliver accurate, contextgrounded clinical summaries.
- Applied machine learning and NLP techniques to clean, normalize, and structure EHR and unstructured clinical text data.
- Performed prompt engineering, retrieval tuning, and system evaluation using retrieval accuracy, response relevance, and latency metrics, and deployed solutions via FastAPI-based production APIs with privacycompliant workflows.
- Containerized the application using Docker and prepared it for scalable cloud deployment.

Outcomes:

- Reduced clinical chart review time by 60% +.
- Improved clinician productivity and faster decision-making.
- Minimized LLM hallucinations through retrieval-grounded responses.
- Enhanced accuracy and completeness of patient history insights.
- Enabled scalable foundation for AI-driven clinical decision support.

Tech Stack:

Python, SQL, Pandas, NumPy, Machine Learning, Natural Language Processing (NLP) Large Language Models (LLMs–Closed Source), Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Vector Databases (Pinecone), Prompt Engineering, Prompt Guardrails, Hallucination Mitigation, Context Grounding, FastAPI REST APIs, Docker, AWS(Secure Cloud), Data Privacy and Compliance (HIPAA-aligned), Model
Evaluation (Retrieval Accuracy, Relevance, Latency)

Enterprise Financial Insights Bot using Multi-Source RAG Project

Duration: 4-5 months | Team Size: 5

Description:

Developed an enterprise-grade Multi-Source Retrieval-Augmented Generation (RAG) financial insights platform to deliver accurate, context-aware financial analysis. The system integrates structured financial data and unstructured documents to support decision-making, risk assessment, compliance, and audit readiness through natural-language queries in a secure enterprise environment.

Roles And Responsibilities:

- Designed and implemented an enterprise-scale Multi-Source RAG system for financial analytics, compliance, and executive decision-support use cases.
- Performed data understanding and preprocessing on structured financial datasets (SQL, CSV) and unstructured sources (PDFs, reports).

- Built scalable data pipelines including chunking, embedding generation, metadata enrichment, and semantic retrieval using vector databases.
- Engineered prompt templates and guardrails to ensure accurate financial reasoning, KPI summaries, and source-grounded LLM responses.
- Integrated closed-source LLMs into production APIs using FastAPI, and evaluated system performance using precision, recall, relevance, and latency metrics.

Outcomes:

- Reduced manual financial analysis and reporting effort by 60% +.
- Improved insight accuracy through grounded, source-based LLM responses.
- Enabled self-service financial querying for business and leadership users.
- Strengthened compliance and audit readiness with traceable explanations.
- Increased operational efficiency by minimizing dependency on manual data interpretation.

Tech Stack:

Python, SQL, Pandas, NumPy, Machine Learning, Natural Language Processing (NLP) Large Language Models (LLMs–Enterprise), Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Vector Databases (Pinecone), Prompt Engineering, Prompt Guardrails, Context Grounding, Chunking Strategies, Metadata Enrichment, FastAPI, REST APIs, Docker, AWS(Secure Cloud), Data Security and Compliance, Model Evaluation (Retrieval Accuracy, Relevance, Latency)

End-to-End Fraud Detection Pipeline for Digital Payments Project
Duration: 4 months | Team Size: 5
Description:

Designed and implemented an end-to-end, production-grade fraud detection pipeline for digital payment transactions to identify high-risk and anomalous activities in near real time. The system analyzes transactional, behavioral, and device-level data to generate fraud risk scores, enabling proactive fraud prevention, reduced financial losses, and improved customer trust while maintaining regulatory compliance.

Roles And Responsibilities:

- Analyzed large-scale digital payment transaction data to identify fraud patterns, anomalies, and high-risk behaviors across multiple channels.
- Performed data cleaning, EDA, and advanced feature engineering on behavioral, velocity, geo-location, and device fingerprinting features.
- Developed and optimized fraud detection models using Logistic Regression, Random Forest, XGBoost, and Isolation Forest with threshold tuning to control false positives.
- Evaluated model performance using Precision, Recall, F1-score, AUC, and KS statistics, aligning model outcomes with business risk objectives.
- Deployed scalable fraud risk scoring services via FastAPI-based REST APIs on cloud infrastructure and monitored performance using drift and stability checks.

Outcomes:

- Improved fraud detection recall by 30% while maintaining controlled false positive rates.
- Improved customer satisfaction by 38%.
- Enabled near real-time fraud risk scoring for transaction blocking and step-up authentication.
- Reduced financial losses through early fraud identification and prevention.
- Decreased manual review effort and operational costs using automated fraud screening.
- Enhanced platform security, customer trust, and regulatory compliance.

Tech Stack:

Python and SQL, Pandas, NumPy, machine learning, Anomaly Detection, Logistic Regression, Random Forest, XGBoost, and Isolation Forest, Feature Engineering, Exploratory Data Analysis (EDA), FastAPI and REST APIs, PostgreSQL/MySQL databases, Docker, Cloud Deployment (AWS), Model Monitoring, Data Drift Detection, Model Evaluation (Precision, Recall, F1-Score, AUC, KS), Power BI, Tableau

CORE COMPETENCIES

- GenAI Application Development
- Machine Learning
- Predictive Modeling
- Deep Learning Architectures
- Time Series & Forecasting
- Model Evaluation & Validation
- Risk & Fraud Analytics

LANGUAGE

- English
- Hindi

EDUCATION

Bachelor of Computer Applications (BCA)
North East Frontier Technical University(NEFTU), Aalo,West Siang District
2019 – 2022